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P102 - A2LA Policy on Measurement Traceability

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Scope and Field of Application¹

The quality of products and services is becoming increasingly dependent on reliable measurements. The importance attached to measurements is reflected in relevant standards by the requirement that measurements must be “traceable” to national or international standards of measurement. Different definitions and explanations of the term “traceability” exist in standards and various literature, giving rise to differing interpretations and misinterpretations.

This policy document is intended to explain the concept of measurement traceability, how it can be achieved, and how it can be demonstrated. A2LA requirements pertaining to measurement traceability are described. This document is intended for all A2LA-accredited and enrolled calibration and testing laboratories, inspection bodies, proficiency testing providers, and reference material producers.

Specific requirements found in this Policy are in ***italic*** type and numbered as in “(T1)”.

A2LA NOTE 1: A separate document (*P113 – A2LA Policy on Measurement Traceability for Life Sciences Testing Laboratories*) has been developed as A2LA’s official interpretation of P102 in the life sciences area specifically. P113 serves as the primary measurement traceability policy for “Life Sciences” testing laboratories (replacing *P102 – A2LA Policy on Measurement Traceability*). It is meant to explain how measurement traceability may be achieved and how it may be demonstrated for biological, environmental and chemical testing laboratories and accomplishes this through creating a mechanism for determining when such materials and calibrations are required by the laboratories in order to meet the reporting requirements of their customers.

A2LA NOTE 2: A separate document (*P102b – Policy on Requesting an Exception to Measurement Traceability*) outlines the requirements to request and receive an exception to the traceability policy.

A2LA NOTE 3: Separate documents (*P905 – A2LA Metrological Traceability Policy for ISO 15189 Laboratory Testing* and *P705 – A2LA Metrological Traceability Policy for CLIA/ISO 15189 Laboratory Testing*) have been developed as A2LA’s official interpretation of P102 in the clinical laboratory testing (ISO 15189 and CLIA) area. P905 and P705 serve as the primary measurement traceability policies for clinical testing laboratories (replacing *P102 – A2LA Policy on Measurement Traceability*).

Definition of terms:

Calibration is defined as: Operation that, under specified conditions, in a first step, establishes a relation between the quantity values with measurement uncertainties provided by measurement standards and corresponding indications with associated measurement uncertainties and, in a second step, uses this information to establish a relation for obtaining a measurement result from an

¹ See, for example, EA-4/07, *Traceability of Measuring and Test Equipment to National Standards*.

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indication (VIM 2.39).

VIM NOTE 1 A calibration may be expressed by a statement, calibration function, calibration diagram, calibration curve, or calibration table. In some cases, it may consist of an additive or multiplicative correction of the indication with associated measurement uncertainty.

VIM NOTE 2 Calibration should not be confused with adjustment of a measuring system, often mistakenly called “self-calibration”, nor with verification of calibration.

VIM NOTE 3: Often, the first step alone in the above definition is perceived as being calibration.

Introduction- the Concept of “Traceability”

Traceability is the property of a measurement result whereby the result can be related to a reference through a documented unbroken chain of calibrations, each contributing to the measurement uncertainty². The purpose of requiring traceability is to ensure that measurements are accurate representations of the specific quantity subject to measurement, within the uncertainty of the measurement.

Traceability is characterized by six essential elements³:

1. *an unbroken chain of comparisons*: going back to stated references acceptable to the parties, usually a national or international standard;
2. *measurement uncertainty*: the uncertainty of measurement for each step in the traceability chain must be calculated or estimated according to agreed methods and must be stated so that an overall uncertainty for the whole chain may be calculated or estimated;
3. *documentation*: each step in the chain must be performed according to documented and generally acknowledged procedures; and the results must be recorded;
4. *competence*: the laboratories or bodies performing one or more steps in the chain must supply evidence for their technical competence (e.g. by demonstrating that they are accredited);
5. *reference to SI units*: the chain of comparisons must, where possible, end at primary standards for realization of the SI units;
6. *calibration intervals*: calibrations must be repeated at appropriate intervals; the length in of these intervals will depend on a number of variables (e.g. uncertainty required, frequency of use, way of use, stability of equipment). See R205: *Specific Requirements: Calibration Laboratory Accreditation Program* for more information .

(T1) A2LA requires that:

(a) All calibrations and verifications of measuring and test equipment and reference standards be

² ILAC P-10:01/2013, *Traceability of Measurement Results*.

³ *ibid*

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conducted by:

- *A calibration laboratory accredited to ISO/IEC 17025 by a mutually recognized Accreditation Body; or,*
- *A recognized National Metrology Institute (NMI) including designated institutes. Recognition of the NMI is based on the Institute or designated institute being a signatory to the CIPM (Comité International des Poids et Mesures) MRA (Mutual Recognition Arrangement) and supporting the measurement comparison activities of the CIPM. A listing of these recognized Institutes can be found at <http://www.bipm.org/en/cipm-mra/participation/signatories.html>; or,*
- *A testing laboratory accredited by A2LA to ISO/IEC 17025 for dimensional testing and found to meet R205: Specific Requirements: Calibration Laboratory Accreditation Program. An accredited test report that meets R205 and R105 – Requirements When Making Reference to A2LA Accredited Status can be considered to satisfy traceability requirements; or,*
- *A laboratory accredited by A2LA to ISO/IEC 17025 and found to meet the T9 requirements of this document for their in-house calibrations; or,*
- *A State Weights and Measures facility with a current certificate of measurement traceability. Please see <http://ts.nist.gov/WeightsAndMeasures/statelabcontact.cfm> for a copy of current certificates.*

(b) When possible, all reference materials shall be obtained from:

- *A reference material producer accredited to ISO Guide 34 by a recognized Asia Pacific Laboratory Accreditation Cooperation (APLAC) signatory recognized for accrediting reference material producers; or,*
- *A recognized National Metrology Institute (NMI) or designated institute.*

(T2) A2LA requires that:

(a) For those external calibrations and verifications performed by an A2LA Accredited calibration laboratory or a calibration laboratory accredited by an MRA partner, these must be recorded in a calibration certificate or report and must include:

1. *An endorsement by the recognized Accreditation Body's symbol (or otherwise makes reference to accredited status by a specific, recognized accreditation body);*
2. *The A2LA certificate number for A2LA-accredited organizations with every use of*

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the A2LA Accredited symbol (or narrative reference) (see R105, Section 1.4.4). and,

3. The measurement uncertainty.

(b) For those external calibrations and verifications performed by an NMI, these must be recorded in a calibration certificate or report and must include:

- 1. An endorsement by the National Metrology Institute (NMI); and,*
- 2. The measurement uncertainty.*

(c) For internal calibrations and verifications, those requirements outlined in requirement T9 of this document apply.

(d) For external calibrations of test equipment requiring traceability through reference materials, these reference materials must be recorded in a certificate meeting the requirements of ISO Guide 31 and must also include:

- 1. an endorsement by the recognized Accreditation Body's symbol (or otherwise makes reference to accredited status by a specific, recognized accreditation body); and,*
- 2. the A2LA certificate number for A2LA-accredited reference material producers (see R105, Section 1.4.4).*

NOTE: *It is common practice for a reference material producer to package their reference materials under a different organization's name. In these instances, it is possible for the reference materials to meet the A2LA Traceability Policy if the accompanying certificate includes reference to the specific, recognized accreditation body and the accreditation certificate number.*

(T3) *All A2LA-Accredited and enrolled organizations must define their policy for achieving measurement traceability and also for achieving traceability for reference materials if applicable. The policy shall ensure compliance with this policy document.*

The Distinction between Calibration and Testing and Achieving Traceability through Dimensional Testing Laboratories

Calibration is defined as the “operation that, under specified conditions, in a first step, establishes a relation between the quantity values with measurement uncertainties provided by measurement standards and corresponding indications with associated measurement uncertainties and, in a second step, uses this information to establish a relation for obtaining a measurement result from an indication”⁴. A test is defined as a “determination of one or more characteristics of an object of conformity assessment, according to a procedure.”⁵

⁴ *International Vocabulary of Basic and General Terms in Metrology (VIM), Definition 2.39.*

⁵ *ISO/IEC 17000-2004, Definition 4.2.*

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In short, “calibration” means determining and documenting the deviation of the indication of a traceable measuring instrument (or the stated value of a material measure) from the conventional “true” value of the measurand. The term “traceability” means a process whereby the indication of a measuring instrument (or a material measure) can be compared, in one or more stages, with a national or international standard for the measurand in question.

Traceability is typically achieved through calibration services. However, in some instances, traceability can be achieved through test results. For example, since A2LA enforces the same requirements on dimensional testing laboratories (including traceability requirements and requirements pertaining to the calculation and reporting of measurement uncertainty) as it does for dimensional calibration laboratories, the distinction between calibration and testing can be lost.

Therefore, a mechanical testing laboratory performing dimensional testing that issues an accredited test report or certificate containing appropriate statements of measurement results, measurement uncertainty, and traceability, in accordance with the requirements of **ISO/IEC 17025:2005 Section 5.10** and the **A2LA Calibration Program Requirements** can be considered as having produced a “calibration” report or certificate for the dimensional artifact in question regardless of the title of the document. This can be particularly useful for complex dimensional artifacts that most dimensional calibration laboratories will not be accredited to calibrate. See *R205: Specific Requirements: Calibration Laboratory Accreditation Program, Appendix D* for more information.

Scopes of Accreditation

Scopes of accreditation are documents that define specifically the measurements an organization is accredited to make. In addition, the scope defines the ranges of the accredited measurand along with the associated best measurement capability expressed as an uncertainty for each measurand and range.

Before placing work with an accredited organization, it is important that the customer request a copy of the organization’s scope (*not the certificate of accreditation*) so that the customer can ensure that the organization is accredited to perform the needed measurements. In addition, customers must ensure that the organization’s measurement uncertainties are suitable for their needs.

Organizations are not permitted to claim a Calibration and Measurement Capability (CMC) on their scope of accreditation that is smaller than the CMC claimed by the National Metrology Institute (as stated in the key comparison database listed on the *Bureau International des Poids et Mesures* [BIPM] website, www.bipm.org) through which traceability is achieved unless allowance is made by A2LA. A2LA may accept uncertainties smaller than the NMI’s “commercial” uncertainty that is provided to its own customers on a case-by-case basis.

Accredited Calibration and Test Reports

For the purpose of demonstrating measurement traceability, calibration certificates shall, wherever

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applicable, indicate the traceability to national or international standards of measurement and should provide the measurement result and associated uncertainty of measurement.

Wherever applicable, and when suitable for customer requirements, a statement of compliance with an identified method or procedural specification can be accepted instead of measurement results and associated uncertainties.

Only calibration certificates or reports endorsed by a recognized accreditation body's symbol (or which otherwise makes reference to accredited status by a specific, recognized accreditation body) that is accompanied by an indication of the type of entity accredited (e.g., "calibration laboratory", "reference material producer") are considered to satisfy traceability requirements. By definition, such an endorsed certificate or report will contain an appropriate statement of measurement results and/or a statement of compliance with an identified metrological specification accompanied by an appropriately defined uncertainty statement and a suitable statement of traceability.

Determination and Statement of Uncertainty

A crucial element of the concept of measurement traceability is measurement uncertainty.

(T4) Where measurement uncertainty analysis is applicable⁶, A2LA requires laboratories to calculate measurement uncertainty in accordance with the ISO "Guide to the Expression of Uncertainty in Measurement." These uncertainties shall be reported as the expanded uncertainty with a defined coverage factor, k (typically $k = 2$) and the confidence interval (typically to approximate the 95% confidence level).

(T5) If a calibration certificate or test report contains a statement of the measurement result and the associated uncertainty, then the uncertainty statement must be accompanied by an explanation of the meaning of the uncertainty statement. An example of such an explanation might be the statement "Reported uncertainties represent expanded uncertainties expressed at approximately the 95% confidence level using a coverage factor of $k = 2$ ". Statements of uncertainty which do not specify at least the coverage factor and the confidence level are incomplete and they are inadequate for the purpose of demonstrating that measurement traceability has been achieved.

It is often the case that a calibration certificate will contain the statement "in tolerance", or words to that effect, along with a statement to the effect that the measurement uncertainty does not exceed a certain fraction of the tolerance. Such fractions are often called "test uncertainty ratios", TURs for short. Uncertainty statements phrased in terms of TURs are implicit statements of the uncertainty: knowing the tolerance ratio allows one to determine the largest possible value of the measurement uncertainty. However,

⁶ Measurement uncertainty analysis is required for all calibrations and dimensional inspections. For applicability of testing, please see the *P103 - Policy on Estimating Measurement Uncertainty for Testing Laboratories* and the relevant Annexes *P103a-P103e*.

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(T6) TURs must be calculated using the expanded uncertainty of the measurement, not the “collective uncertainty of the measurement standards”⁷.

(T7) Implicit uncertainty statements must be accompanied by words to the effect that the uncertainty ratio was calculated using the expanded measurement uncertainty. In addition the coverage factor and confidence level must be stated.

Statements of Traceability

This statement will affirm that the calibration reported was conducted using standards whose values are traceable to an appropriate national, international, intrinsic, or mutual consent standard. For example, if the traceability chain for a given laboratory originates at NIST, then the statement will affirm that “This calibration was conducted using standards traceable to the SI through NIST”, or words to that effect.

Calibration certificates and reports which do not contain equivalent statements of traceability, or which only refer to NIST report of test numbers as evidence of traceability are insufficient to demonstrate measurement traceability.

(T8) In addition to the information required in the above sections, calibration reports and certificates must contain a traceability statement⁸.

⁷ This is the language in ANSI/NCSLI Z540-1-1994 section 10.2.b.

⁸ See ANSI/NCSLI Z540-1-1994, section 13.2.q.

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In-house Calibrations⁹

An in-house calibration is an internal calibration, not listed on the organization's Scope of Accreditation, of CAB owned measuring and test equipment (M&TE).

An in-house calibration system ensures that all measuring and test equipment used in a company is calibrated regularly against its own reference standards. The in-house calibration system shall ensure traceability of measurement by having its reference standards calibrated at an accredited calibration laboratory or a national metrology institute.

The nature and scope of the metrological control of in-house calibration is at the discretion of the parent organization. They must be adapted to the particular applications so that the results obtained with the measuring and test equipment are sufficiently accurate and reliable. Accreditation of organizations to perform in-house calibrations is not always necessary to satisfy the requirements of ISO/IEC 17025. However, in light of the definition of measurement traceability,

(T9) all in-house calibrations must be supported by the following minimal set of elements:

- a) The in-house laboratory must maintain documented procedures for the in-house calibrations.***
- b) The in-house calibrations must be evidenced by a calibration report, certificate, or sticker, or other suitable method;***
- c) Calibration records must be retained for an appropriate, prescribed time;***
- d) The in-house laboratory must maintain training records for calibration personnel and these records must demonstrate the technical competence of the personnel performing the calibrations: evidence of competence includes, for example, documented training and the results of measurement audits;***
- e) The in-house laboratory shall be able to demonstrate traceability to national or international standards of measurement by procuring calibration services from appropriately accredited calibration labs or an NMI for the reference standards and/or purchasing reference materials from appropriately accredited reference material producers or an NMI;***
- f) The in-house laboratory must have and apply procedures for evaluating measurement uncertainty. Measurement uncertainty shall be calculated in accordance with the GUM for each type of calibration and records of these calculations shall be maintained. Measurement uncertainty must be taken into account when statements of compliance with specifications are made.***

⁹ See EA-4/07, *Traceability of Measuring and Test Equipment to National Standards*.

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- g) Reference standards must be recalibrated at appropriate intervals to ensure that the reference value is reliable. Policy and procedures for establishing and changing calibration intervals must be based on the historical behavior of the reference standard¹⁰.***

It is not always easy to define the precise circumstances under which a given calibration should be considered to be an in-house calibration that is not subject to accreditation requirements. However at least two cases can be distinguished:

- 1) If the calibration service is performed within the same physical location as the customer¹¹, and if the calibrations are performed in a permanent calibration laboratory (i.e., customer equipment to be calibrated is transported to the calibration laboratory), then the calibration should be considered to be an in-house calibration not subject to accreditation requirements;
- 2) If the calibration service is performed at a location other than a permanent calibration laboratory (i.e., if reference standards are being transported to the customer equipment to be calibrated), then such a calibration service should be accredited.

Acceptable Accreditors of Calibration and Testing Providers

A2LA has signed multi-lateral recognition agreements or arrangements (MLAs) with numerous accreditation bodies throughout the world. The import of these agreements is that the signatories promote the recognition and acceptance of certificates and reports issued by organizations accredited by accreditation bodies who have signed the MLA. Through the vehicle of the MLA, a uniform level of competence of the accredited bodies involved is assured and the need for multiple assessments is diminished or eliminated. This means that a supplier should only need one certificate or report to satisfy the markets and governments represented by MLA signatories.

Currently, the primary multi-lateral recognition agreements amongst accrediting bodies are the Asia-Pacific Laboratory Accreditation Cooperation (APLAC¹²), the International Laboratory Accreditation Cooperation (ILAC¹³) and the Inter-American Accreditation Cooperation (IAAC¹⁴). As signatories to these multi-lateral agreements, A2LA is committed to promoting the recognition and acceptance of accreditations granted by its fellow signatories.

Accredited test and calibration results, reported by laboratories that are accredited by the accreditation

¹⁰ See, for example, NCSLI RP-1 "Establishment & Adjustment of Calibration Intervals" (3/96).

¹¹ "customer" refers to the recipient of the in-house calibration service.

¹² <http://www.aplac.org>

¹³ <http://www.ilac.org/>

¹⁴ <http://iaac.org.mx>

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bodies recognized by any of these multi-lateral agreements, and reported in a test or calibration report endorsed by the accrediting body's symbol (or which otherwise makes reference to accredited status by a specific, recognized accreditation body, for example through use of a statement that the organization is accredited by XYZ) that is accompanied by an indication of the type of entity accredited (e.g., through inclusion of an accreditation certificate number, words such as "calibration laboratory", etc.), are recognized by A2LA as satisfying the requirements pertaining to measurement traceability.

A2LA recognizes reference material certificates that are issued by reference material producers that are accredited by the accreditation bodies recognized by the APLAC mutual recognition arrangement for reference material producer accreditation, and reported in a certificate meeting ISO Guide 31 and endorsed by the accreditation body's symbol (or which otherwise makes reference to accredited status by a specific, recognized accreditation body) and an indication of the type of entity accredited.

Use of NIST Test Report Numbers as Evidence of Traceability

The NIST Calibration Program¹⁵ often receives calls to verify the authenticity of a NIST Report of Test numbers appearing on another organization's report. Although NIST can verify the authenticity of its report numbers, having an authentic number does not provide assurance or evidence that the measurement value provided by another organization is traceable. Not only should there be an unbroken chain of comparisons, each measurement should be accompanied by a statement of uncertainty associated with the farthest link in the chain from NIST, that is, the last facility providing the measurement value. NIST does not have that information; only the facilities that provided the measurement values to the customer can provide the associated uncertainties and describe the traceability chain.

To establish an audit trail for traceability, a proper calibration result should include: the assigned value, a stated uncertainty, identification of the standards used in the calibration, and the specification of any environmental conditions of the calibration where correction factors should be applied, if the standard or equipment were to be used under different environmental conditions.

Similarly, it is the policy of the National Conference of Standards Laboratories International (NCSLI) that test report numbers issued by NIST are intended to be used solely for administrative purposes. Although they are often used to uniquely identify documents which bear evidence of traceability, test report numbers shall not be used nor required as proof of the adequacy or traceability of a test or measurement¹⁶.

It should also be noted that nationally and internationally recognized standards dealing with test and

¹⁵http://www.nist.gov/traceability/nist_traceability_policy_external.cfm.

¹⁶ NCSLI Position Statement 96-1.

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measurement quality requirements such as ANSI/NCSL Z540-1, ISO 10012, ISO/IEC 17025 and the ISO9000 series do not require the use or reporting of NIST test report numbers to establish traceability.

Consequently, A2LA neither requires nor accepts the presence of NIST test report numbers on test or calibration reports as sufficient evidence of the traceability of a measurement result.

Summary of Specific Requirements

(T1) A2LA requires that:

(a) All calibrations and verifications of measuring and test equipment and reference standards, be conducted by:

- A calibration laboratory accredited to ISO/IEC 17025 by a mutually recognized Accreditation Body; or,
- A recognized National Metrology Institute (NMI) including designated institutes. Recognition of the NMI is based on the Institute or designated institute being a signatory to the CIPM (Comité International des Poids et Mesures) MRA (Mutual Recognition Arrangement) and supporting the measurement comparison activities of the CIPM. A listing of these recognized Institutes can be found at <http://www.bipm.org/en/cipm-mra/participation/signatories.html>; or,
- A testing laboratory accredited by A2LA to ISO/IEC 17025 for dimensional testing and found to meet R205: Specific Requirements: Calibration Laboratory Accreditation Program. An accredited test report that meets R205 and R105 – Requirements When Making Reference to A2LA Accredited Status can be considered to satisfy traceability requirements; or,
- A laboratory accredited by A2LA to ISO/IEC 17025 and found to meet the T9 requirements of this document for their in-house calibrations; or,
- A State Weights and Measures facility with a current certificate of measurement traceability. Please see <http://ts.nist.gov/WeightsAndMeasures/statelabcontact.cfm> for a copy of current certificates.

(b) When possible, all reference materials shall be obtained from:

- A reference material producer accredited to ISO Guide 34 by a recognized Asia Pacific Laboratory Accreditation Cooperation (APLAC) signatory recognized for accrediting reference material producers; or,

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- A recognized National Metrology Institute (NMI) or designated institute.

(T2) A2LA requires that:

(a) For those external calibrations and verifications performed by an A2LA Accredited calibration laboratory or a calibration laboratory accredited by MRA partner, these must be recorded in a calibration certificate or report and must include:

1. An endorsement by the recognized Accreditation Body's symbol (or otherwise makes reference to accredited status by a specific, recognized accreditation body);
2. The A2LA certificate number for A2LA-accredited organizations with every use of the A2LA Accredited symbol (or narrative reference) (see R105, Section 1.4.4); and,
3. The measurement uncertainty.

(b) For those external calibrations and verifications performed by an NMI, these must be recorded in a calibration certificate or report and must include:

1. An endorsement by the National Metrology Institute (NMI); and,
2. The measurement uncertainty.

(c) For internal calibrations and verifications, those requirements outlined in requirement T9 of this document apply.

(d) For external calibrations of test equipment requiring traceability through reference materials, these reference materials must be recorded in a certificate meeting the requirements of ISO Guide 31 and must also include:

1. an endorsement by the recognized Accreditation Body's symbol (or otherwise makes reference to accredited status by a specific, recognized accreditation body); and,
2. the A2LA certificate number for A2LA-accredited reference material producers (see R105, Section 1.4.4).

NOTE: It is common practice for a reference material producer to package their reference materials under a different organization's name. In these instances, it is possible for the reference materials to meet the A2LA Traceability Policy if the accompanying certificate includes reference to the specific, recognized accreditation body and the accreditation certificate number.

(T3) All A2LA-Accredited and enrolled organizations must define their policy for achieving measurement traceability and also for achieving traceability for reference materials if applicable. The policy shall ensure compliance with this policy document.

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(T4) Where measurement uncertainty analysis is applicable, A2LA requires laboratories to calculate measurement uncertainty in accordance with the ISO “Guide to the Expression of Uncertainty in Measurement.” These uncertainties shall be reported as the expanded uncertainty with a defined coverage factor, k (typically $k = 2$) and the confidence interval (typically to approximate the 95% confidence level).

(T5) If a calibration certificate or test report contains a statement of the measurement result and the associated uncertainty, then the uncertainty statement shall be accompanied by an explanation of the meaning of the uncertainty statement. (For example, “This uncertainty represents an expanded uncertainty expressed at approximately the 95% confidence level using a coverage factor of $k=2$.”)

(T6) TURs shall be calculated using the expanded uncertainty of the measurement, not the “collective uncertainty of the measurement standards”.

(T7) Implicit uncertainty statements must be accompanied by words to the effect that the uncertainty ratio was calculated using the expanded measurement uncertainty. In addition the coverage factor and confidence level must be stated.

(T8) Calibration reports and certificates issued by A2LA-accredited calibration laboratories shall contain a traceability statement.

(T9) All in-house calibrations must be supported by the following minimal set of elements:

- (a) The in-house laboratory must maintain documented procedures for the in-house calibrations;
- (b) The in-house calibrations must be evidenced by a calibration report, certificate, or sticker, or other suitable method;
- (c) Calibration records must be retained for an appropriate, prescribed time;
- (d) The in-house laboratory must maintain training records for calibration personnel and these records must demonstrate the technical competence of the personnel performing the calibrations. Evidence of competence includes, for example, documented training and the results of measurement audits;
- (e) The in-house laboratory shall be able to demonstrate traceability to national or international standards of measurement by procuring calibration services from appropriately accredited calibration labs or an NMI and/or purchasing reference materials from appropriately accredited reference material producers or an NMI;
- (f) The in-house laboratory must have and apply procedures for evaluating measurement uncertainty. Measurement uncertainty shall be calculated for each type of calibration and records of these calculations shall be maintained. Measurement uncertainty must be taken into account when statements of compliance with specifications are made;

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- (g) Reference standards must be recalibrated at appropriate intervals to ensure that the reference value is reliable. Policy and procedures for establishing and changing calibration intervals must be based on the historical behavior of the reference standard.

APPENDIX A - Document Revision History

Date	Description
6/29/2012	- Added T2 (d-3). - Clarified T2 (a-2) and T2 (d-2) to match P101 requirements for traceable certificates. - Added definition of “in house calibration”.
5/9/2013	- Updated reference to ILAC P10:01/2013 - Updated A2LA NOTE 1 to include <i>P113 – A2LA Policy on Measurement Traceability for Life Sciences Testing Laboratories</i> - Added A2LA NOTE 3 - Updated web links for APLAC and IAAC
5/7/2014	- Editorial changes as a result of A2LA P101 change to R105 (clauses T1.a., T2.a.2., T2.d.2). - T1.a, 2nd bullet, deleted ‘R205c’ - T1.b, 1st bullet, deleted ‘in combination with ISO/IEC 17025’ - Added Annex P103e to footnote 6 - References to P605 updated to P905/P705.